Notes: Helpman, Melitz & Yeaple (AER, 2004)
Export Versus FDI with Heterogeneous Firms

Contents

1 Big picture

- **Question.** Why do some firms serve foreign markets through exports while others use horizontal foreign direct investment, and why does the export-to-FDI ratio differ across industries and countries?
- **Core mechanism.** Firms differ in productivity. Serving foreign consumers by exports requires lower fixed costs but higher variable trade costs. Serving them by FDI requires higher fixed costs but lower variable costs.
- **Sorting prediction.** The least productive firms serve only the domestic market; more productive firms export; the most productive firms undertake FDI.
- **Central cross-industry prediction.** Industries with more dispersed firm productivity should have relatively more affiliate sales and fewer exports, because the upper tail contains more very productive firms that find it profitable to pay the fixed cost of FDI.
- **Empirical setting.** The paper compares U.S. exports and foreign affiliate sales by industry and destination country, using proxies for freight costs, tariffs, plant-level fixed costs, and firm-size dispersion.
- **Main empirical result.** Higher firm-size dispersion is associated with lower U.S. exports relative to affiliate sales, even after controlling for proximity-concentration variables.
- **Economic takeaway.** The organization of international trade depends not only on country characteristics and trade costs, but also on the distribution of firm productivity within industries.

2 Incremental contribution of the paper

1. **Combines heterogeneous-firm trade with multinational production.** Melitz-style firm heterogeneity explains export selection; Helpman-Melitz-Yeaple extend it to the choice between exporting and FDI.
2. **Gives microfoundations to the proximity-concentration trade-off.** Earlier horizontal FDI models focused on trade costs versus plant fixed costs. This paper shows that firm productivity dispersion changes the aggregate export-versus-FDI outcome.
3. **Generates a new cross-industry prediction.** Conditional on freight costs, tariffs, and plant fixed costs, industries with greater firm-size dispersion should have relatively more FDI sales.
4. **Links firm productivity sorting to aggregate trade composition.** The paper connects firm-level productivity ranks to country-industry export/FDI ratios.
5. **Introduces multiple empirical measures of dispersion.** It constructs dispersion measures from U.S. Census data and European Amadeus firm-level data, then shows that results are robust across them.
6. **Shows that firm heterogeneity is quantitatively comparable to standard proximity-concentration forces.** The beta coefficients show that dispersion has an economic impact similar to freight, tariffs, and plant fixed costs.

Interpretation: The incremental contribution is not simply adding another determinant of FDI. The paper changes the unit of analysis: aggregate export/FDI patterns are explained by the shape of the within-industry firm productivity distribution.

3 Data and measurement

3.1 International sales measures

The dependent variable is the log of U.S. exports relative to U.S. foreign affiliate sales for country-industry pairs. The numerator is U.S. export sales to destination country j in industry h. The denominator is sales by

U.S. majority-owned foreign affiliates in the same destination and industry. The data are for 1994.

Interpretation: A lower export-to-FDI ratio means more activity is served by local production through affiliates rather than by shipping goods from the United States.

3.2 Proximity-concentration variables

- **FREIGHT:** ratio of CIF imports to FOB imports, measuring transport and insurance costs.
- **TARIFF:** ad valorem tariff measure from BEA country-industry data.
- **FP:** proxy for plant-level fixed costs, based on average nonproduction workers per establishment.

Interpretation: The proximity-concentration prediction is: high freight and tariffs make exports less attractive relative to FDI, while high plant-level fixed costs make FDI less attractive relative to exports.

3.3 Firm heterogeneity measures

Because productivity is not observed for all firms, the paper uses the dispersion of firm sales as a proxy for the dispersion of productivity. Under a Pareto productivity distribution, the dispersion of firm sales maps into the shape parameter of productivity. The paper uses five measures:

- U.S. standard deviation of log sales;
- Europe standard deviation of log sales;
- France standard deviation of log sales;
- Europe regression coefficient from log rank on log sales;
- France regression coefficient from log rank on log sales.

Interpretation: The use of multiple dispersion measures is important. The U.S. measure is semiaggregated, while the European and French measures use actual firm-level sales data. Similar results across measures support the interpretation that dispersion captures a real industry property.

3.4 Controls and samples

The paper uses a narrow sample of 27 countries and a wider sample that adds 11 smaller or developing countries. Regressions include country dummies, and some specifications include capital intensity and R&D intensity controls, random effects, clustered standard errors, and IV strategies for dispersion.

4 Model and theoretical specification

4.1 Firm options

After paying an entry cost, a firm draws a productivity level $1/a$. It can choose among three foreign-market modes:

1. stay domestic only;
2. export to foreign consumers;
3. establish a foreign affiliate and serve the market through local production.

Profits from each mode are increasing in productivity, but they have different fixed and variable cost structures.

4.2 Profit functions

In simplified notation, operating profits can be represented as:

$$\pi_D(a) = a^{1-\varepsilon}B - f_D,$$

$$\pi_X(a) = (\tau a)^{1-\varepsilon}B - f_X,$$

$$\pi_I(a) = a^{1-\varepsilon}B - f_I.$$

Here $\tau > 1$ is the iceberg trade cost, f_X is the fixed cost of exporting, and f_I is the fixed cost of FDI. FDI avoids trade costs but requires higher fixed costs.

Interpretation: Because $\varepsilon > 1$, more productive firms have lower a and earn larger variable profits. The steepness and intercepts of the profit lines generate cutoffs for domestic production, exporting, and FDI.

4.3 Sorting

There are productivity cutoffs such that:

- low-productivity firms exit or serve only the domestic market;
- intermediate-productivity firms export;
- high-productivity firms undertake FDI.

Interpretation: This is the core sorting prediction. The model explains why multinationals are more productive than exporters and why exporters are more productive than purely domestic firms.

4.4 Cross-industry implication

Let s_X^{Uj}/s_I^{Uj} denote the relative export-to-affiliate sales ratio from the United States to country j. The theoretical expression implies:

$$\frac{s_X^{Uj}}{s_I^{Uj}} = \left(\omega^j \tau_h^{Uj}\right)^{1-\varepsilon_h} \left[\frac{V_h^U(a_{hX}^{Uj})}{V_h^U(a_{hI}^{Uj})} - 1\right].$$

Under the Pareto benchmark, higher dispersion in firm size lowers the export-to-FDI ratio.

Interpretation: Firm heterogeneity changes aggregate trade composition because a thicker upper tail increases the mass of firms for which FDI is profitable.

5 Identification and empirical design

5.1 Regression target

The empirical specification relates the log of relative exports to proximity-concentration variables and dispersion:

$$\log\left(\frac{Exports_{hj}}{AffiliateSales_{hj}}\right) = \alpha_j + \beta_1 FREIGHT_{hj} + \beta_2 TARIFF_{hj} + \beta_3 FP_h + \beta_4 DISPERSE_h + \Gamma Controls_h + \varepsilon_{hj}.$$

The theory predicts:

- $\beta_1 < 0$: high freight lowers exports relative to FDI;

- $\beta_2 < 0$: high tariffs lower exports relative to FDI;
- $\beta_3 > 0$: high plant fixed costs lower FDI relative to exports;
- $\beta_4 < 0$: greater productivity dispersion raises FDI relative to exports.

5.2 What identifies the dispersion effect

The effect is identified by cross-industry variation in firm-size dispersion, compared across country-industry observations, after controlling for country fixed effects and standard proximity-concentration variables.

Interpretation: The empirical design is not a natural experiment. It tests whether the cross-sectional pattern predicted by the model holds after controlling for alternative explanations.

5.3 Main threats and responses

- **Measurement error in dispersion.** The paper uses five dispersion measures and instruments the U.S. dispersion measure with European measures.
- **Omitted sector characteristics.** It adds capital intensity and R&D intensity controls and random industry effects.
- **Country heterogeneity.** Country dummies absorb destination-specific fixed export and FDI costs.
- **Interdependence across countries.** Robustness specifications aggregate Europe and cluster standard errors by industry.
- **Alternative proximity-concentration forces.** Freight, tariffs, and plant fixed costs are included directly.

Interpretation: The strongest evidence is that dispersion remains negative and significant across samples, measurement strategies, and robustness checks.

6 Tables and figures

6.1 Table 1: Productivity Advantage of Multinationals and Exporters; Figure 1: Profits from Domestic Sales, from Exports, and from FDI

Read: Table 1 shows that multinationals are more productive than nonmultinational exporters, and both groups are more productive than the remaining firms. Figure 1 shows the theoretical cutoff structure for domestic-only firms, exporters, and multinational firms.

Economic message: Table 1 provides preliminary firm-level support for the model's sorting mechanism. Figure 1 explains why the sorting arises: exporting has lower fixed costs but FDI has lower variable costs, so the most productive firms prefer FDI.

TABLE 1—PRODUCTIVITY ADVANTAGE OF MULTINATIONALS AND EXPORTERS

Multinational	0.537
	(14.432)
Nonmultinational exporter	0.388
	(9.535)
Coefficient difference	0.150
	(3.694)
Number of firms	3,202

Notes: T-statistics are in parentheses (calculated on the basis of White standard errors). Coefficients for capital intensity controls and industry effects are suppressed.

(a) **Table 1: Productivity Advantage of Multinationals and Exporters.** Multinationals are about 15 percentage points more productive than nonmultinational exporters after controls.

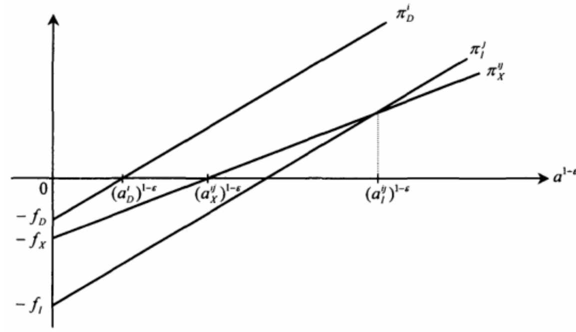

FIGURE 1. PROFITS FROM DOMESTIC SALES, FROM EXPORTS, AND FROM FDI

(b) **Figure 1: Profits from Domestic Sales, from Exports, and from FDI.** The figure shows the profit cutoffs that sort firms into domestic-only, export, and FDI modes.

6.2 Figure 2: Empirical Distribution of Firm Sales; Figure 3: Regression Fit to the Pareto Distribution

Read: Figure 2 plots log firm rank against log sales in four sectors. Figure 3 plots estimated Pareto slopes and confidence intervals across sectors.

Economic message: The figures validate the empirical strategy of using firm-size dispersion as a proxy for productivity dispersion. If firm sales distributions are approximately Pareto, the slope of rank-size relationships can proxy for heterogeneity.

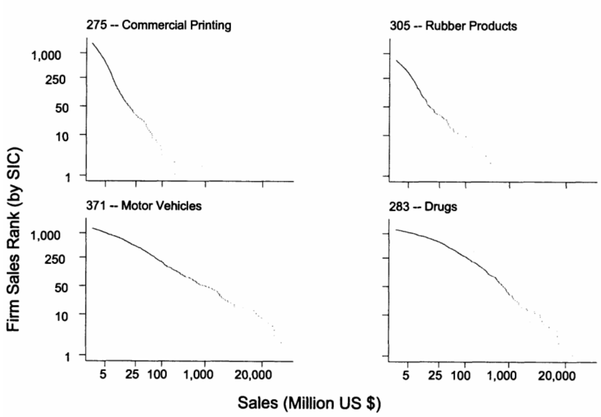

FIGURE 2. EMPIRICAL DISTRIBUTION OF FIRM SALES

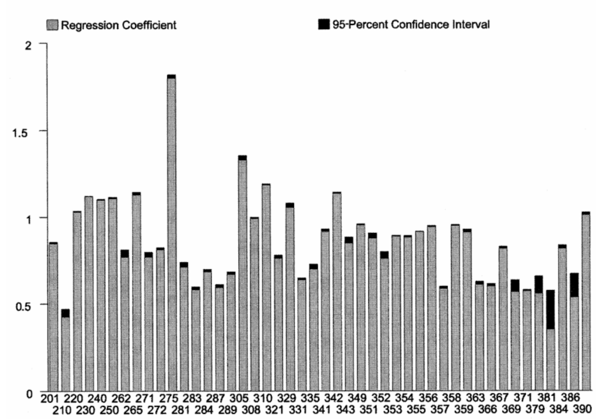

FIGURE 3. REGRESSION FIT TO THE PARETO DISTRIBUTION

(a) **Figure 2: Empirical Distribution of Firm Sales.** Rank-size plots are close to linear on log-log scales, consistent with Pareto tails.

(b) **Figure 3: Regression Fit to the Pareto Distribution.** Regression slopes by sector quantify cross-industry differences in dispersion.

6.3 Table 2: Correlations between Alternative Measures of Dispersion; Table 4: "Beta" Coefficients

Read: Table 2 reports that the European and French dispersion measures are highly correlated with each other. The U.S. measure is positively correlated but less tightly aligned. Table 4 translates regression coefficients into standardized beta coefficients.

Economic message: Table 2 supports the use of multiple dispersion proxies. Table 4 shows that the economic magnitude of dispersion is comparable to freight costs, tariffs, and plant-level fixed costs.

TABLE 2—CORRELATIONS BETWEEN ALTERNATIVE MEASURES OF DISPERSION

	U.S. std. dev.	Europe std. dev.	France std. dev.	Europe reg. coeff.	France reg. coeff.	FP	RD	KL
U.S. s.d.	1							
Europe s.d.	0.507	1						
France s.d.	0.567	0.895	1					
Europe reg.	0.526	0.959	0.919	1				
France reg.	0.541	0.973	0.905	0.984	1			
FP	0.455	0.621	0.508	0.652	0.624	1		
RD	0.134	0.445	0.354	0.438	0.475	0.498	1	
KL	0.129	0.585	0.500	0.507	0.523	0.515	0.365	1

(a) **Table 2: Correlations between Alternative Measures of Dispersion.** Most European dispersion measures are highly correlated, suggesting they capture a common industry property.

TABLE 4—"BETA" COEFFICIENTS: NARROW SAMPLE WITH CONTROLS

	Mean	Standard deviation	"Beta" coefficient
Dependent variable	−0.595	2.375	
FREIGHT	1.863	0.653	−0.271
TARIFF	2.015	1.020	−0.205
FP	3.321	0.785	0.325
U.S. s.d.	1.749	0.316	−0.312
Europe s.d.	1.198	0.276	−0.250
France s.d.	1.224	0.375	−0.325
Europe reg.	1.260	0.333	−0.210
France reg.	1.257	0.336	−0.211

(b) **Table 4: Beta Coefficients.** Standardized magnitudes show that dispersion has explanatory power similar to standard proximity-concentration variables.

6.4 Table 3: Exports Versus FDI; Table 5: Exports Versus FDI - Random Effects

Read: Table 3 reports the baseline OLS results. Freight and tariffs have negative coefficients, fixed plant costs have positive coefficients, and all dispersion measures have negative coefficients. Table 5 adds industry random effects and capital/R&D controls.

Economic message: These are the main empirical tests. The negative dispersion coefficients mean that industries with more firm heterogeneity have lower exports relative to affiliate sales, as predicted.

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TABLE 3—EXPORTS VERSUS FDI

Narrow sample (N = 961)	U.S. std. dev.	Europe std. dev.	France std. dev.	Europe reg. coeff.	France reg. coeff.
FREIGHT	−1.040	−0.959	−1.019	−0.935	−0.944
	(−7.392)	(−6.749)	(−7.328)	(−6.526)	(−6.594)
TARIFF	−0.365	−0.512	−0.421	−0.545	−0.539
	(−2.644)	(−3.636)	(−3.917)	(−3.781)	(−3.775)
FP	1.177	0.932	0.927	0.947	0.934
	(10.159)	(7.827)	(8.059)	(7.453)	(7.450)
DISPERSE	−2.343	−2.153	−2.061	−1.503	−1.491
	(−8.374)	(−5.250)	(−6.664)	(−4.535)	(−4.470)
KL	−0.868	−0.495	−0.456	−0.628	−0.626
	(−7.790)	(−4.529)	(−4.256)	(−5.876)	(−5.859)
RD	−0.104	0.007	0.007	0.006	−0.002
	(−2.197)	(0.150)	(0.144)	(0.125)	(−0.047)
R^2	0.373	0.340	0.364	0.332	0.334

Wide sample (N = 1,175)	U.S. std. dev.	Europe std. dev.	France std. dev.	Europe reg. coeff.	France reg. coeff.
FREIGHT	−1.011	−0.935	−0.960	−0.915	−0.919
	(−7.968)	(−7.246)	(−7.714)	(−7.040)	(−7.053)
TARIFF	−0.241	−0.384	−0.306	−0.411	−0.407
	(−1.876)	(−2.964)	(−2.457)	(−3.073)	(−3.057)
FP	1.133	0.861	0.868	0.867	0.848
	(10.428)	(7.719)	(7.994)	(7.318)	(7.243)
DISPERSE	−2.248	−1.866	−1.833	−1.284	−1.215
	(−8.611)	(−4.919)	(−5.982)	(−4.132)	(−3.924)
KL	−0.793	−0.454	−0.412	−0.569	−0.576
	(−7.483)	(−4.347)	(−3.982)	(−5.574)	(−5.636)
RD	−0.086	0.017	0.021	0.015	0.007
	(−1.914)	(0.367)	(0.446)	(0.326)	(0.153)
R^2	0.338	0.305	0.325	0.298	0.298

(a) **Table 3: Exports Versus FDI.** Baseline regressions support both the proximity-concentration trade-off and the firm-heterogeneity prediction.

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TABLE 5—EXPORTS VERSUS FDI—RANDOM EFFECTS

Narrow sample (N = 961)	U.S. std. dev.	Europe std. dev.	France std. dev.	Europe reg. coeff.	France reg. coeff.
FREIGHT	−0.430	−0.398	−0.428	−0.397	−0.397
	(−2.554)	(−2.344)	(−2.533)	(−2.336)	(−2.334)
TARIFF	−0.113	−0.127	−0.105	−0.136	−0.133
	(−0.922)	(−1.033)	(−0.857)	(−1.107)	(−1.085)
FP	1.376	1.132	1.096	1.154	1.137
	(5.145)	(4.128)	(4.233)	(4.107)	(4.093)
DISPERSE	−2.623	−2.763	−2.445	−2.031	−1.991
	(−4.897)	(−3.459)	(−4.761)	(−3.098)	(−3.180)
KL	−1.106	−0.613	−0.570	−0.757	−0.758
	(−4.652)	(−2.238)	(−2.168)	(−2.891)	(−2.896)
RD	−0.002	0.126	0.116	0.133	0.119
	(−0.020)	(1.029)	(0.970)	(1.081)	(0.972)
R^2	0.352	0.316	0.342	0.307	0.308

Wide sample (N = 1,175)	U.S. std. dev.	Europe std. dev.	France std. dev.	Europe reg. coeff.	France reg. coeff.
FREIGHT	−0.331	−0.322	−0.328	−0.320	−0.320
	(−2.296)	(−2.230)	(−2.278)	(−2.215)	(−2.215)
TARIFF	−0.004	−0.018	−0.004	−0.022	−0.021
	(−0.038)	(−0.155)	(−0.035)	(−0.193)	(−0.187)
FP	1.361	1.110	1.081	1.127	1.103
	(4.123)	(3.455)	(3.475)	(3.377)	(3.369)
DISPERSE	−2.518	−2.559	−2.265	−1.864	−1.786
	(−3.824)	(−2.733)	(−3.706)	(−2.398)	(−2.424)
KL	−1.069	−0.599	−0.561	−0.734	−0.739
	(−3.660)	(−1.871)	(−1.789)	(−2.373)	(−2.408)
RD	0.006	0.123	0.116	0.129	0.115
	(0.042)	(0.862)	(0.811)	(0.894)	(0.805)
R^2	0.312	0.274	0.297	0.267	0.267

(b) **Table 5: Exports Versus FDI - Random Effects.** Dispersion remains negative and significant after adding random effects and sector controls.

6.5 Table 6: Exports Versus FDI - Additional Robustness Results

Read: Table 6 uses clustered standard errors and IV specifications. The IV specifications instrument the U.S. dispersion measure with European dispersion measures.

Economic message: The key dispersion coefficient remains negative and statistically significant. This reduces the concern that the baseline result is driven by measurement error in U.S. dispersion or by unobserved sectoral dependence.

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TABLE 6—EXPORTS VERSUS FDI—ADDITIONAL ROBUSTNESS RESULTS
(Clustered standard errors and IV specifications)

	OLS			IV		
	Narrow sample	Wide sample	Aggregated Europe	Narrow sample	Wide sample	Aggregated Europe
FREIGHT	−1.040	−1.011	−1.001	−1.218	−1.118	−1.053
	(−3.997)	(−4.437)	(−4.464)	(−3.862)	(−4.366)	(−4.545)
TARIFF	−0.365	−0.241	−0.077	−0.188	−0.124	−0.016
	(−1.611)	(−1.081)	(−0.304)	(−0.706)	(−0.478)	(−0.056)
FP	1.177	1.133	1.086	1.609	1.457	1.344
	(4.876)	(4.472)	(4.166)	(3.605)	(3.311)	(3.182)
DISPERSE (U.S.)	−2.343	−2.248	−2.150	−4.321	−3.681	−3.198
	(−3.689)	(−3.655)	(−3.349)	(−2.606)	(−2.248)	(−2.217)
KL	−0.868	−0.793	−0.848	−0.938	−0.848	−0.890
	(−3.032)	(−2.513)	(−2.593)	(−2.855)	(−2.496)	(−2.599)
RD	−0.104	−0.086	−0.087	−0.158	−0.127	−0.121
	(−0.851)	(−0.687)	(−0.691)	(−1.233)	(−0.985)	(−0.968)
R^2	0.373	0.338	0.314	0.328	0.315	0.315
N	961	1,175	678	961	1,175	678

Notes: In the IV specifications, the U.S. dispersion measure is instrumented using all four European dispersion measures. All *T*-statistics (in parentheses) are computed from standard errors that are heteroskedasticity consistent and adjusted for clustering by industry. Constant and country dummies are suppressed.

Table 6: Exports Versus FDI - Additional Robustness Results. The negative relationship between dispersion and the export-to-FDI ratio survives clustering, alternative samples, and IV strategies.

7 Short summary

- Firms differ in productivity and sort into domestic-only, exporting, and FDI modes.
- Exporting has lower fixed costs but higher variable trade costs.
- FDI has higher fixed costs but lower variable costs.
- The most productive firms choose FDI; less productive internationally active firms export.
- Industries with more productivity dispersion have more high-productivity firms and therefore more FDI relative to exports.
- The paper measures dispersion using U.S., European, and French firm-sales distributions.
- Regression evidence supports the model: high freight and tariffs reduce exports relative to FDI, high plant fixed costs raise exports relative to FDI, and high dispersion reduces exports relative to FDI.
- Dispersion has economically large effects comparable to traditional proximity-concentration variables.

8 Conclusion

8.1 Core conclusion

Helpman, Melitz, and Yeaple show that the export-versus-FDI decision depends not only on trade costs and plant fixed costs, but also on within-industry productivity heterogeneity. A standard proximity-concentration trade-off explains why firms substitute affiliate production for exports when trade costs are high or plant-level fixed costs are low. The paper's key addition is that the distribution of productivity determines how many firms are productive enough to choose FDI.

8.2 Economic intuition

The intuition is simple but powerful. Exporting is cheaper to start but costly at the margin because goods must be shipped across borders. FDI is expensive to start but cheaper at the margin because it avoids trade costs. Low-productivity firms cannot pay either foreign-market fixed cost. Moderately productive firms export. Highly productive firms choose FDI because they sell enough abroad to justify the higher fixed cost of a foreign affiliate.

8.3 Incremental contribution revisited

The paper brings together two strands of trade theory: horizontal FDI theory based on proximity-concentration trade-offs and heterogeneous-firm trade theory based on productivity selection. The resulting model predicts not only which firms serve foreign markets, but also how industry-level firm heterogeneity changes the aggregate mix of exports and multinational activity.

8.4 Empirical implication

The empirical implication is that industries with more dispersed firm-size distributions should have lower export-to-FDI ratios. The data support this prediction across multiple dispersion measures and robustness tests. This makes the paper a foundational contribution to the empirical literature on multinational firms and heterogeneous producers.

8.5 Limitations

The empirical analysis is cross-sectional and uses proxies for productivity dispersion, fixed costs, and trade costs. Dispersion measures are imperfect, and some sector characteristics may remain unobserved. The paper addresses these concerns with multiple dispersion measures, controls, random effects, clustered standard errors, and IV strategies, but the design is not a natural experiment.

8.6 Takeaway

The paper's main lesson is that internationalization is not a single margin. Firms first select into foreign markets and then choose how to serve those markets. The distribution of firm productivity is central to both margins and therefore to the observed structure of world trade and multinational production.